Customer Oriented Software Quality

Introduction

- Software developer and SQA Engineers - accept that a common definition of quality does not exist

- Companies – decided to ask their customers to define quality

How do your customer's define quality?

The Quality Attributes Set

- **Quality attribute set** is a way to represent customer quality requirements

- To create QAS
  - Ask your customers
  - By consolidating their responses , develop the set of quality attributes that describes quality requirements

- A customer oriented SQA program is designed to satisfy customer quality requirements
- Example:
  - COSQA(Customer oriented software quality assurance) Company

Building the QAS(Quality Attribute Set)

- Following are the two ways to ask customer about their quality requirements
  - Interview
  - Questionnaire

The Quality Attributes Set

- Product Specific Attributes
  - Ease of Use
    - Important attribute
    - Many Manufacturers mention it in their advertisements
    - COSQA Software's customers prefer to purchase
      - Products that do not require them to read the manual
      - Products that do not use the on-line help facilities
      - Products with GUI (Look and feel)
      - They can Learn the products by playing with it

The Quality Attributes Set

- Documentation
  - There can be incomplete ,missing, inaccurate, and poorly understood documentation
  - Customer conclude that a product is of poor quality
  - According to COSQA Software's customers
    - Expect all documentation to be online
    - Comprehensive documentation

The Quality Attributes Set

- Defect tolerance
  - Defect is an undesirable behavior or characteristic of a product
  - Some defects are more severe than others
  - Any example?
  - Customers will not tolerate defects that alter or destroy their work in progress or that adversely affect their productivity.
- Defect frequency
  - Defect ---tolerable—Occur with greater frequency?

The Quality Attributes Set

- Defect impact
  - If a defect results in a long operational delay, its effect may not be negligible.
  - Example:
    - Large telemarketing company –relies upon a DBMS to record orders, make inquiries…
    - Each minute the database is down represents $1,000,000 of lost revenue.
- Packaging
  - Customers believed that quality products are found only inside attractive packages.

The Quality Attributes Set

- Price versus Reliability
  - How could a products 's price possibly affect quality?
- Performance
  - It does not necessarily refer to how fast or how slow it operates
  - Instead it refers to whether or not the product lives up to customer expectations.
  - Customers rely on trade journals and magazines to provide them with accurate information

The Quality Attributes Set

Organization –Specific attributes

- Service and Support
  - Customers will need help from you directly
  - Don’t like automated phone system
- Internal processes
  - The quality of an organization is also judged by the process/standard followed by that organization